llmXive Automated Review of MulTaBench: Benchmarking Multimodal Tabular Learning with Text and Image

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses strictly on the accuracy of factual claims and the validity of their supporting citations.

The manuscript makes several strong claims regarding the state of the art and the properties of existing benchmarks. In the Introduction, the authors state that “ConTextTab set the SOTA for the CARTE benchmark” citing [spinaci_contexttab_2025]. A check of the provided bibliography reveals a potential mismatch or missing entry for this specific citation key (the .bib file lists `arazi_tabstar_2025` but the text cites `spinaci_contexttab_2025` in multiple places, and the specific SOTA claim on CARTE needs to be explicitly verifiable in the cited work). If the cited paper does not explicitly claim SOTA status on CARTE, this is an unsupported factual assertion.

In Section 3, the selection of embedding models is justified by citing [muennighoff_mteb_2023] for “high performance-to-parameter efficiency.” This citation key is missing from the provided `neurips2026.bib` file. While the claim about the models’ efficiency might be true, the citation provided does not exist in the bibliography, rendering the support for this specific claim invalid in the current manuscript state.

Furthermore, in Section 5, the text discusses the comparison between “TAR Small” and “Frozen Large” variants. The sentence structure implies that the observation that TAR Small outperforms Frozen Large is a finding of the current study, which is correct. However, the sentence immediately preceding it cites [grinsztajn_vectorizing_2023] regarding the standard practice of PCA. The flow of the paragraph risks conflating the citation of standard practice with the novel empirical finding. The authors should ensure the citation is not misread as supporting the specific “TAR Small > Frozen Large” result, which is a new contribution of this paper.

Finally, the Abstract claims MulTaBench is the “largest image-tabular benchmarking effort to date.” This is a superlative claim that requires rigorous verification against all cited benchmarks (MuG, BAG, TIME, etc.). The text mentions MuG has 4 data sources and BAG has 11, but the authors should explicitly confirm that no other un-cited or partially cited benchmark exceeds the 20-image dataset count to ensure the claim is not an overstatement.

These issues are primarily regarding the precision of citations and the verification of superlative claims. They do not invalidate the core science but require correction to ensure factual accuracy and proper attribution.

Action items.

- **[writing]** The review focuses strictly on the accuracy of factual claims and the validity of their supporting citations. The manuscript makes several strong claims regarding the state of the art and the properties of existing benchmarks. In the Introduction, the authors state that “ConTextTab set the SOTA for the CARTE benchmark” citing [spinaci_contexttab_2025]. A check of the provided bibliography reveals a potential mismatch or missing entry for this specific citation key (the .bib file lists arazi_tabs

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively compares model performance across conditions, but the right panel is missing Y-axis tick labels, hindering quantitative interpretation. Additionally, the legend terminology slightly diverges from the caption's phrasing.

Figure 2

The figure presents normalized scores with confidence intervals but suffers from a legend that obscures data points in panel (a). Additionally, the distinction between the 'E2E' (purple) and 'Frozen/TAR' (blue/orange) categories is not explained in the caption, leaving the selection criteria for these groups ambiguous.

Figure 3

The figure is visually clear with a defined legend, but the 'Curation' legend entry is misleading regarding the star symbol's placement, and the caption lacks necessary context on the normalization baseline and metric.

Figure 4

Figure 4 presents computation costs with a clear legend and caption, but the logarithmic scaling on the runtime plot appears visually inconsistent with the bar heights, and the separation line's alignment with x-axis categories could be clearer.

Figure 5

The figure presents a comparison of tabular learners using different encoders, but the title and caption misleadingly frame this as an 'Encoder Scale Analysis' rather than a model comparison. Additionally, the x-axis lacks a descriptive label.

Figure 6

The figure effectively visualizes the performance of different models with error bars, but the caption's description of 'normalized within each model' conflicts with the strictly positive 0-1 axis range, and the y-axis lacks a formal title.

Figure 7

The figure effectively communicates the ablation results with clear error bars and a distinct legend, but the legend title 'PCA / mode' is semantically inaccurate as it groups 'No PCA' entries under a PCA header.

Figure 8

The figure displays a raw X-ray image but fails to show the attention map overlay described in the caption, making the claim about attention shifts unverifiable.

Figure 9

The figure is a photograph of a dog that lacks the promised attention map overlay, and the caption incorrectly references 'cat ears' which are not present in the image.

Figure 10

The figure is a raw retinal image that fails to display the 'Attention Maps' described in the caption. Without visual overlays or comparative panels showing the attention distributions, the figure does not support the claims regarding the difference between Frozen and TAR methods.

Figure 11

The figure is a raw photograph that fails to display the attention maps described in the caption, making it impossible to verify the claims regarding 'Frozen' versus 'TAR' focus areas.

Figure 12

Figure 12 effectively visualizes the curation pipeline described in the caption. The flowchart clearly depicts the sequential filtering steps (Joint Frozen vs. Structured/Unstructured, then TAR vs. Joint Frozen) and the corresponding discard conditions, making the inclusion criteria for MulTaBench immediately understandable.

Action items.

- [**science**] Figure 1: The right panel (MulTaBench) lacks a visible Y-axis with tick labels, making it impossible to read specific normalized scores or compare magnitudes against the left panel.
- [**writing**] Figure 1: The caption refers to 'Structured' and 'Unstructured' conditions, but the legend uses 'Unimodal Structured' and 'Unimodal Unstructured'; the legend should be updated to match the caption for consistency.
- [**science**] Figure 2: The legend defines 'E2E' (purple) and 'Curation' (star), but the bars for AG-MM, TabSTAR, and ConTextTab are purple without stars, while others are blue/orange with stars. The caption does not explain the criteria for a model to be classified as 'E2E' versus 'Frozen/TAR', nor does it explain why some models lack the 'Curation' star despite

being in the benchmark.

- [**writing**] Figure 2: The legend box is positioned over the data bars in panel (a), partially obscuring the error bars and the 'Curation' star for the 'TabST' and 'RealMLP' rows, reducing readability.
- [**science**] Figure 3: The legend defines 'Curation' with a star symbol, but the star appears on the left side of the bars (indicating a specific model variant or condition) rather than as a separate data point or error bar, making the legend definition misleading.
- [**writing**] Figure 3: The caption claims to show 'Normalized scores over MulTaBench' but does not specify the normalization baseline (e.g., 0 = random, 1 = oracle) or the metric used (e.g., R2, MSE), which is essential for interpreting the 'Normalized Score' axis.
- [**science**] Figure 4: The left panel's y-axis is logarithmic (10^3, 10^4), but the bar heights for 'Image Small' and 'Text Small' appear visually inconsistent with the log scale (e.g., the 'Image Small' Frozen bar is near the bottom, but on a log scale starting at 10^2, it should be much higher relative to the 10^3 line).
- [**writing**] Figure 4: The x-axis labels ('Image Small', 'Image Large', etc.) are split across two lines, which is acceptable, but the spacing between the 'Image' and 'Text' groups is not clearly demarcated by the dashed line's position relative to the labels, potentially causing confusion about which category the line separates.
- [**science**] Figure 5: The y-axis labels (LightGBM, CatBoost, etc.) represent tabular learners, but the figure title and caption claim to analyze 'Encoder Scale' (Small vs. Large). The figure actually compares different tabular models using different encoders, rather than analyzing the scale of a single encoder across models, creating a disconnect between the visual data and the stated analysis goal.
- [**writing**] Figure 5: The x-axis label 'Normalized Score' is missing; while the axis ticks are present, the unit/metric name is not explicitly labeled on the axis itself.
- [**science**] Figure 6: The x-axis is labeled 'Normalized Score' and ranges from 0 to 1, but the caption states scores are 'normalized within each model'. This typically implies centering (mean=0) or scaling to a [-1, 1] range, yet the data is strictly positive and bounded by 1, suggesting a different normalization (e.g., min-max) that is not explicitly defined in the caption.
- [**writing**] Figure 6: The y-axis labels (LightGBM, CatBoost, etc.) are positioned to the left of the plot area but lack a clear axis title (e.g., 'Model') to formally identify the categorical variable.
- [**writing**] Figure 7: The legend title 'PCA / mode' is semantically confusing because the entries describe both PCA status ('No PCA') and dimensionality ('30 dims'); the title should be changed to 'Configuration' or 'Method' to accurately reflect the content.
- [**writing**] Figure 7: The x-axis labels ('CatBoost', 'LightGBM') are centered between the two groups of bars, which is acceptable, but the spacing is tight; ensuring the labels are

clearly associated with their respective clusters would improve readability.

- [**science**] Figure 8: The image is a raw chest X-ray without any visible attention map overlay (e.g., heatmap or gradient visualization) to support the caption's claim that 'attention shifts from diffused edges to the lung'.
- [**writing**] Figure 8: The caption references 'attention shifts' but the image lacks a legend or color scale to interpret the attention intensity or distribution.
- [**fatal**] Figure 9: The image displays a puppy, but the caption claims 'Attention isolates the cat ears and the dog's eyes.' The image contains no cat, making the caption factually incorrect and the figure's scientific claim unverifiable.
- [**science**] Figure 9: The image is a standard photograph with no visible attention map overlay (e.g., heatmap or saliency mask), failing to visually demonstrate the 'Attention Maps' described in the title and caption.
- [**fatal**] Figure 10: The rendered image is a standard fundus photograph of a retina, but the caption describes 'Attention Maps' comparing 'Frozen' vs 'TAR' methods. The image lacks the necessary visual overlays (e.g., heatmaps) or side-by-side panels to demonstrate the claimed attention shifts.
- [**science**] Figure 10: The caption claims to show 'Frozen attention scatters randomly' and 'TAR converges,' but the image contains no visual indicators (such as color gradients or masks) to represent these attention distributions, making the scientific claim unverifiable from the figure.
- [**science**] Figure 11: The image displays a single photograph of a person without any overlaid attention heatmaps or visualizations. This contradicts the caption's claim that it shows 'Attention Maps' comparing 'Frozen' and 'TAR' conditions, rendering the figure unable to support its scientific claim.
- [**science**] Figure 11: The filename in the caption ('idx31_label1_original.png') suggests a raw input image, but the figure is cited to demonstrate model behavior (attention focus). The figure fails to visualize the actual data (attention weights) described in the text.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and technical shorthand that impede accessibility for a broader machine learning audience. The most critical issue is the introduction of the core concept "Target-Aware Representations" (TAR) in the Abstract without defining the acronym. The term "TAR" is then used repeatedly throughout the paper (e.g., Abstract, Section 1, Section 3, Section 4) without ever being explicitly defined as an acronym in the text, forcing the reader to infer the meaning or search for the definition.

Similarly, “MMTL” (Multimodal Tabular Learning) is introduced in Section 1 without the full phrase being spelled out in the immediate vicinity of the acronym’s first use, and it is subsequently used as a standalone noun phrase. “ICL” (In-Context Learning) appears in Section 2 without definition. While “LoRA” is a standard technique, it is introduced in Section 1 without expansion.

Additionally, the phrase “lossy summaries” in Section 1 uses signal-processing jargon (“lossy”) that could be replaced with “compressed” or “information-reducing” to improve clarity for non-specialists. The paper would benefit from a strict policy of spelling out all acronyms at their first occurrence and replacing niche jargon with plainer alternatives where possible.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific acronyms and technical shorthand that impede accessibility for a broader machine learning audience. The most critical issue is the introduction of the core concept “Target-Aware Representations” (TAR) in the Abstract without defining the acronym. The term “TAR” is then used repeatedly throughout the paper (e.g., Abstract, Section 1, Section 3, Section 4) without ever being explicitly defined as an acronym in the text, forcing the reader to infer t

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning (MMTL), supported by a rigorous curation pipeline. The central premise—that frozen embeddings lose task-specific signal—is well-motivated by the theoretical limitations of general-purpose encoders and empirically validated by the curation results. The conclusion that MulTaBench is a necessary benchmark for developing future Multimodal Tabular Foundation Models follows logically from the finding that existing benchmarks contain a high proportion of datasets where TAR provides no benefit.

However, there are minor logical gaps regarding the generalizability of the results to optimally tuned baselines. The paper demonstrates that TAR outperforms frozen embeddings across a suite of *default* tabular learners. It explicitly states that hyperparameter optimization (HPO) was omitted for the new models to reduce computational cost. While the authors argue this underestimates performance, the logical leap that TAR gains will persist (or widen) when the frozen baselines are also optimally tuned is an assumption, not a proven fact. It is plausible that a heavily tuned frozen embedding pipeline could close the performance gap, potentially weakening the claim that TAR is *universally* superior across all modeling choices. The conclusion should be slightly tempered to specify that TAR gains are robust across *standard/default* configurations.

Additionally, the definition of the benchmark creates a tautological loop regarding the "necessity" of TAR. The benchmark is curated *specifically* to include only datasets where TAR improves performance. While this is a valid design choice for a specialized benchmark, the paper's broader claim that "existing benchmarks mask the benefits of task-specific tuning" relies on the implicit assumption that the datasets rejected by the pipeline (where TAR failed) are not representative of the general MMTL problem space. The paper would benefit from a brief discussion on why the rejected datasets (approx. 60% of the pool) are less relevant for the specific goal of advancing Multimodal TFMs, rather than simply being a different class of problems where frozen embeddings suffice.

Finally, the comparison between "TAR Small" and "Frozen Large" supports the claim that tuning is more critical than scale. However, the logical chain is incomplete without a direct comparison to "TAR Large". While the paper implies that tuning is the dominant factor, the absence of a "TAR Large" baseline leaves a small gap in proving that the *combination* of large scale and tuning is not the true optimum, which could affect the architectural recommendations for future TFMs.

Action items.

- [**writing**] The paper presents a logically coherent argument for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning (MMTL), supported by a rigorous curation pipeline. The central premise—that frozen embeddings lose task-specific signal—is well-motivated by the theoretical limitations of general-purpose encoders and empirically validated by the curation results. The conclusion that MulTaBench is a necessary benchmark for developing future Multimodal Tabular Foundation Models f

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the novelty and scope of MulTaBench and the necessity of Target-Aware Representations (TAR) that slightly exceed the immediate evidence provided.

First, the claim in the Abstract and Introduction that MulTaBench constitutes the "largest image-tabular benchmarking effort to date" is an overreach. While the paper introduces 20 image-tabular datasets, it does not explicitly quantify the total number of datasets or samples in prior benchmarks (e.g., MuG, BAG, or the image subsets of AutoGluon benchmarks) to substantiate the superlative "largest." Without a direct comparative count, this statement is an unsupported extrapolation.

Second, the Introduction asserts that "no existing model has successfully maintained SOTA performance on tabular tasks while learning TAR for text and images." This is a definitive negative claim. The paper supports this by citing TabSTAR's compromise on numerical performance and the limitations of PFNs, but it does not present a direct, controlled ablation where a SOTA tabular model is explicitly forced to learn TAR versus remaining frozen on a unified benchmark to prove the impossibility of the former. The evidence suggests a *trend* or *difficulty*, but the absolute phrasing overstates the conclusion.

Finally, the robustness analysis concludes that TAR outperforms frozen embeddings "even at the larger scale" (Sec 5). While the results for DINO-v3 and e5 are clear, the paper extrapolates this to imply a general property of embedding models. It does not consider that other pretraining objectives or architectures might inherently encode task-relevant signals without fine-tuning. The claim should be tempered to reflect that this finding holds for the specific class of encoders evaluated.

These issues are primarily matters of phrasing and scope definition rather than fundamental scientific flaws, but they require correction to ensure the claims are strictly supported by the data.

Action items.

- [**writing**] The claim that MulTaBench is the 'largest image-tabular benchmarking effort to date' (Abstract, Sec 1) is an overreach. The paper acknowledges existing benchmarks like MuG and BAG but lacks a quantitative comparison of dataset counts or total samples to substantiate the 'largest' superlative. Please add a comparative table or explicit count comparison.
- [**writing**] The assertion that 'no existing model has successfully maintained SOTA performance on tabular tasks while learning TAR' (Sec 1) is too absolute. The paper cites TabSTAR's limitations but lacks a direct, controlled experiment comparing a SOTA model with TAR against one without on a unified benchmark. This strong negative claim requires stronger evidence or softer phrasing (e.g., 'current models struggle to...').
- [**writing**] The conclusion that 'TAR significantly outperforms frozen embeddings even at the larger scale' (Sec 5) overgeneralizes from specific encoders (DINO-v3, e5). The paper does not rule out that other architectures might inherently capture task-relevant signals without fine-tuning. Qualify the claim to reflect it holds for the specific family of encoders evaluated.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses safety and ethics primarily through the use of publicly available, de-identified datasets and the inclusion of a standard NeurIPS ethics checklist. The authors correctly note in the checklist (Item 9) that no human subjects were involved in the study itself and that the data is public. However, given the inclusion of high-stakes domains such as healthcare (CheXpert, CBIS-DDSM, Glaucoma) and social bias (Celeb Attractiveness), the manuscript would benefit from a more explicit discussion of the downstream ethical implications of the proposed benchmark.

Specifically, in Section 1 (Introduction) and Section 7 (Discussion), the authors should briefly address the potential for harm if models trained on MulTaBench are deployed in clinical settings without rigorous validation, particularly regarding the risk of false negatives in medical diagnosis. Additionally, the use of the 'Celeb Attractiveness' dataset, which relies on crowd-sourced subjective ratings, warrants a brief acknowledgment of the ethical concerns regarding bias and the potential for reinforcing harmful stereotypes if such models are used in real-world applications.

While the authors state that the data is de-identified, the inclusion of images of people (CelebA) and animals (PetFinder) requires a confirmation that the release of the benchmark complies with the original data licenses and that reasonable steps have been taken to mitigate re-identification risks, even if the data is public. The current checklist response (Item 11) is somewhat generic; a more specific statement regarding the handling of these specific image datasets would strengthen the ethical review. The paper does not appear to contain dual-use risks that would prevent publication, but these clarifications are necessary for responsible research dissemination.

Action items.

- [**writing**] The paper includes medical datasets (CheXpert, CBIS-DDSM, Glaucoma) but lacks a specific discussion on the ethical risks of deploying models trained on these benchmarks in clinical settings, particularly regarding false negatives in diagnosis.
- [**writing**] The 'Celeb Attractiveness' dataset relies on crowd-annotated attractiveness labels. The manuscript should explicitly address the ethical implications of using such subjective, potentially biased human annotations for training automated systems.
- [**writing**] While the authors state they use de-identified data, the 'Celeb Attractiveness' and 'PetFinder' datasets contain images of identifiable individuals or animals. A brief statement confirming compliance with the original data licenses regarding public release and potential re-identification risks is needed.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling argument for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning (MMTL) and introduces MulTaBench, a curated benchmark of 40 datasets. The experimental design is generally robust, utilizing 5 random seeds, multiple tabular learners, and ablation studies on encoder scale and dimensionality. The evidence supporting the central claim—that frozen embeddings lose task-specific signal—is strong, with consistent performance gains observed across the majority of models and datasets.

However, the scientific validity of the benchmark's curation criteria warrants closer scrutiny. The definition of "Task-awareness" is operationally defined by the success of a specific algorithmic intervention (LoRA finetuning of the last 3 layers). This creates a potential circularity: datasets are selected *because* they respond to this specific intervention, which inherently biases the benchmark toward models that utilize similar finetuning strategies. While the authors acknowledge this entanglement in the Discussion, the paper does not sufficiently rule out the possibility that the observed "Task-awareness" is an artifact of the specific encoder architecture (DINO/e5) or the finetuning hyperparameters rather than an intrinsic property of the data distribution.

Furthermore, the methodological choice to discretize regression targets into 20 bins for the TAR finetuning step (Appendix, Section "Regression") introduces a potential confound. There is no analysis provided to demonstrate that this discretization preserves the fine-grained signal necessary for the downstream regression task, nor is there a comparison against direct regression finetuning. If the discretization smooths out the very noise or fine-grained details that TAR is supposed to recover, the validity of the "Task-awareness" claim for regression tasks is weakened.

Finally, while the paper claims generalization across learners, the win rates for non-curation models (e.g., TabICLv2 at 55%, ConTextTab at 73%) are notably lower than for the curation models (80-93%). This suggests the benchmark may be somewhat specialized to the inductive biases of the models used during curation. The authors should provide a more nuanced discussion on the extent to which the benchmark results are transferable to architectures with fundamentally different learning dynamics.

Action items.

- [**science**] The curation pipeline entangles 'Task-awareness' with the specific LoRA finetuning algorithm. Datasets are selected because TAR works, creating a circular bias. Clarify if this property is intrinsic to the data or an artifact of the specific encoder/finetuning strategy used for selection.
- [**science**] Discretizing regression targets into 20 bins for TAR finetuning may distort the signal. Provide evidence that this preserves fine-grained information or compare against direct regression finetuning to rule out this confounding variable.
- [**science**] Non-curation models (e.g., TabICLv2, ConTextTab) show significantly lower win rates (55-76%) than curation models. Discuss whether the benchmark is overfit to the specific inductive biases of the curation models and if generalization claims hold for other architectures.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis presented in MulTaBench is generally robust in its experimental design, utilizing multiple random seeds (n=5) and diverse tabular learners to establish the superiority of Target-Aware Representations (TAR) over frozen embeddings. The use of 95% confidence intervals in the main results figures (e.g., Figure 3, Figure 4) correctly conveys the variability across seeds. However, several critical statistical details are missing or require clarification to ensure full reproducibility and validity of the claims.

First, the method for calculating the reported 95% confidence intervals is not explicitly defined in the text or appendices. It is unclear whether these intervals represent the standard error of the mean (SEM), the standard deviation (SD) of the performance metric, or a bootstrap-derived interval. Furthermore, the assumption of normality required for standard parametric confidence intervals is not verified, which is particularly relevant given the small number of seeds (n=5) used for aggregation. The authors should explicitly state the formula or method used (e.g., "95% CI calculated as mean ± 1.96 * SEM assuming normal distribution") to allow for proper interpretation of the error bars.

Second, the curation pipeline described in Section 3.2 and Appendix A.3 relies on a binary decision rule: a dataset is accepted only if the performance gain (Delta) exceeds a fixed threshold (delta=0.001) for at least 3 out of 5 models. This approach introduces a potential selection bias. Datasets with high variance in performance across seeds might fail the threshold due to random fluctuation rather than a lack of true "Joint Signal" or "Task-awareness." Conversely, datasets with low variance but small effect sizes might pass. The paper does not discuss the statistical power of this selection mechanism or the probability of Type I/II errors in the curation process. A sensitivity analysis varying the threshold (delta) or the consensus fraction (rho) would strengthen the robustness of the benchmark selection.

Third, the handling of regression targets in the TAR fine-tuning step (Appendix A.1) involves discretizing continuous labels into 20 equal-frequency bins and optimizing for cross-entropy. While the authors state this is "more stable," they provide no statistical comparison to a direct regression fine-tuning objective (e.g., MSE or MAE). Discretization inherently discards information and introduces quantization error. Without an ablation study showing that the discretized approach yields comparable or superior downstream performance (R^2) compared to direct regression fine-tuning, this methodological choice remains an unverified assumption that could impact the validity of the TAR gains reported for regression tasks.

Finally, while the paper reports win rates in Table 10 of the Appendix, it does not provide statistical significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) to confirm that the observed win rates (e.g., 91.5% for CatBoost) are significantly different from chance (50%) or from each other. Given the large number of dataset-model pairs, even small effect sizes could be statistically significant, but the current presentation lacks this formal inference.

Action items.

- **[writing]** Clarify the statistical method used to compute the 95% confidence intervals reported in Figures 3, 4, 5, and 6. The text mentions '95% CIs over all runs' but does not specify if these are standard error of the mean (SEM), standard deviation (SD), or bootstrap intervals, nor does it state the distributional assumptions (e.g., normality) used for the calculation.

- **[science]** The curation pipeline (Section 3.2) uses a binary acceptance threshold (delta=0.001) and a consensus rule (3/5 models). The statistical power of this selection process is not discussed. There is a risk of selection bias where datasets with high variance might be rejected or accepted based on random seed fluctuations rather than true signal. A sensitivity analysis or power calculation for the curation threshold is needed.
- **[science]** In Appendix A.1, regression targets are discretized into 20 bins for the TAR fine-tuning step. The paper claims this is 'more stable' but does not provide statistical evidence (e.g., comparison of MSE or R^2) that this discretization preserves the predictive signal better than direct regression fine-tuning. This methodological choice requires justification or an ablation study.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with a clear narrative arc and well-structured arguments. The introduction effectively motivates the need for Target-Aware Representations (TAR) using the pneumonia example, and the transition from theoretical desiderata to the curation pipeline is logical. The prose is generally concise, and the distinction between "Joint Signal" and "Task-awareness" is articulated with precision.

However, there are minor issues with sentence flow and clarity in specific sections that, while not obscuring the meaning, could be polished for a top-tier venue. In Section 5, the statistical comparison regarding the acceptance rate of image-tabular datasets ("a proportion comparable to the text-tabular subset") is slightly ambiguous regarding the base numbers used for the percentage, which could confuse a careful reader. Additionally, in Section 6, a few sentences are overly dense, combining multiple complex ideas (e.g., the performance of ConTextTab, the SOTA status on CARTE, and the implications for MulTaBench) into a single clause, which slightly hampers readability.

The Appendix contains a few instances of minor grammatical awkwardness, such as redundant prepositions or slightly clunky phrasing in the description of the trimodal dataset analysis. These do not constitute errors but represent opportunities to tighten the prose. Overall, the writing is strong and effectively communicates the technical contributions, requiring only light editing to achieve perfect clarity.

Action items.

- **[writing]** In Section 5 (Image-Tabular Curation), the phrase 'from which only 5 meet our criteria (31%), a proportion comparable to the text-tabular subset' is ambiguous. It is unclear if the 31% refers to the 5/16 or the final 20/56. Clarify the denominator to ensure the comparison is mathematically precise.

- **[writing]** In Section 6 (Robustness Analysis), the sentence 'This finding is particularly telling, as ConTextTab has set the SOTA for the CARTE Benchmark, emphasizing that MulTaBench targets a fundamentally different text-tabular problem' is slightly convoluted. Consider splitting this into two sentences to improve flow and emphasize the contrast between the benchmarks.
- **[writing]** In the Appendix (Text-Image-Tabular Datasets), the phrase 'By applying the selection rule independently, we prove that the 3 modalities contribute to the prediction to fulfill the Joint Signal criterion' contains a redundant 'to'. Rephrase to '...contribute to the prediction, thereby fulfilling the Joint Signal criterion' for better readability.